

Course: 2024/2025 Data Base Bachelor's in Data Science and Engineering, Dual Bachelor's in Data Science and Engineering and Telecommunication Technologies Engineering. SUBJECT: Project 1 (Part 2: Queries DB)	 Universidad Carlos III de Madrid
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Statement

Given the practical case design solution "Online Chess Platform", answer the following questions based on the relational schema/diagram and the following semantic assumptions.

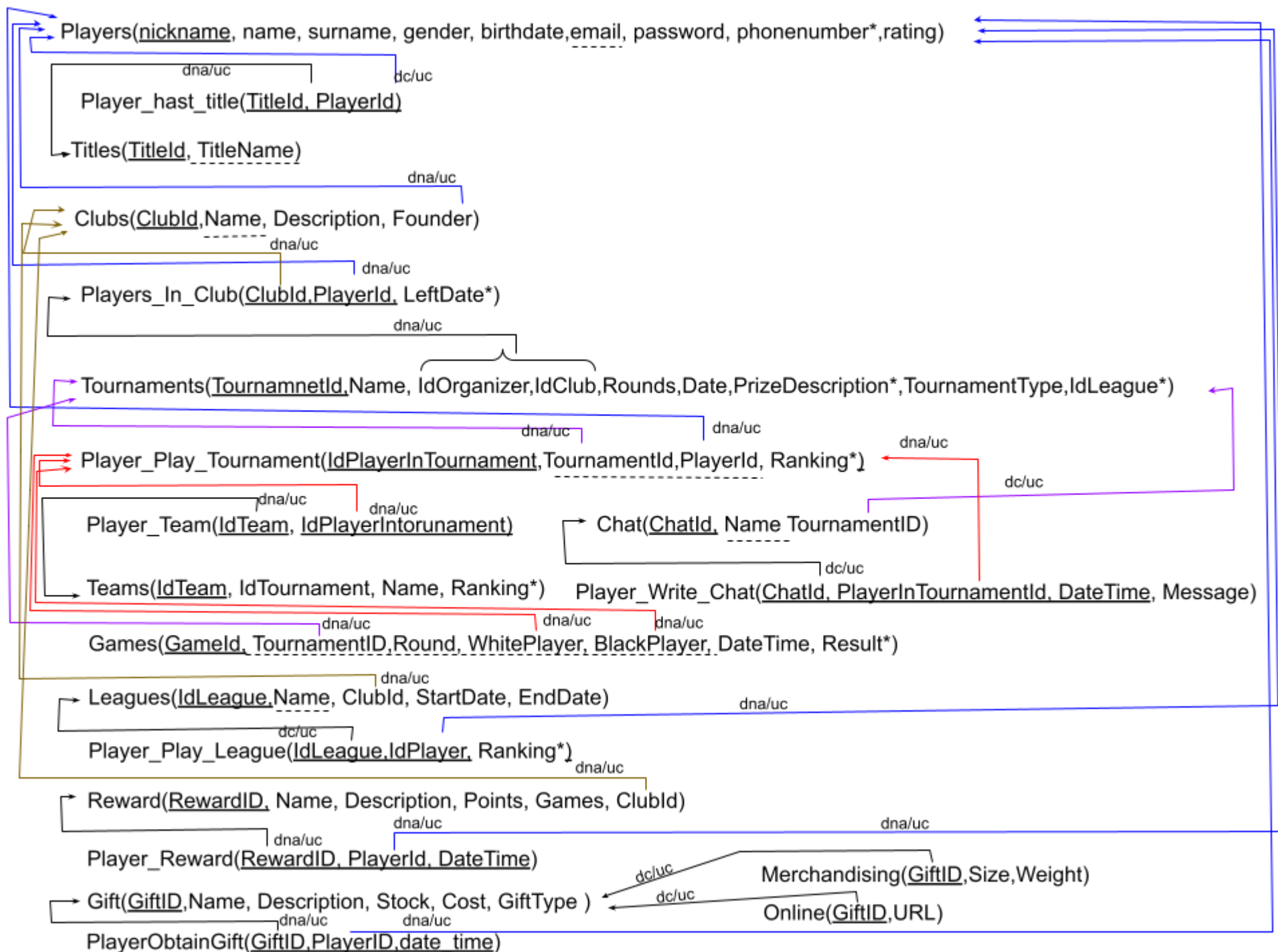
Assumptions to the schema

- The default value for Player.rating is 1500.
- On creating a chess club the database will automatically include the founder as a club player (trigger).
- If a player is back to the club after leaving it before, the date of left is set to null (no history of enrollments).
- The founder cannot left the club. To let the founder left the club it is necessary to change the FounderId of the club before.
- Players who left the club cannot play tournaments organized by this club.
- The maximum size of Tournaments.PrizeDescription is 200 characters
- The players of a game must be registered in the corresponding tournament. We use the relation with Player instead of Player_In_Club because in this way in the future it eases to allow players from other clubs (or without being part of a club) play tournaments.
- Players receive an Id when they play a tournament. That id is the primary key of the table Player_play_Tournament.
- The combination of TournamentId and PlayerId is AK of the table Player_Play_Tournament.
- In the same round of the same tournament the players of a game cannot play another game.
- The round number in the table Game must be consistent with the maximum number of rounds of the tournament (for example, if a tournament have 5 rounds a game of that tournament cannot have the number 6).
- In the same tournament, two teams cannot have the same name.
- When the result of a game is updated, the database will update the rating of the game's players.
- When the result of a game is updated, the database will check the rewards for the winner and update player's rating.
- A tournament can be part of only one league.
- The players must be signed up in the tournament related to a chat to send messages.
- The league and the tournament must be organized by the same club.
- The player signed up in a league must be part of the club who organises it (LeftDate null). We use the relation with Player instead of Player_In_Club because in this way in the future it eases to allow players from other clubs (or without being part of a club) play leagues.
- At the end of the league the database will calculate the ranking by summing the scores obtained in all the tournaments of the league.
- The player who obtains a reward must belong to the club that offers the reward (trigger).

- The database must control that the gift information is stored in the right table (for example if the gift type is "Online" the GiftID cannot be inserted in the table Merchandising) .
- When the player acquires a gift the database automatically update the stock
- If the player doesn't have enough points to get a gift, the database will cancel the transaction (trigger).
- When a player wants to acquire a gift, If there isn't enough stock of the gift the database will reject the acquisition (trigger)
- Domains:
 - Player.gender: M, F
 - Player.rating. Integer value ≥ 0
 - Tournament.Rounds ≥ 5
 - Toruntaments.Type = teams, individual.
 - Game.Result: 0,1,-1
 - Gift.Type = "Merchandising", "Online"

Assumptions to the statement

- Rating will always be a positive number or 0.
- Two titles cannot have the same title name.
- The founder must be included as a club player.
- If a player is back to the club after leaving it before, the date of left is set to null (no history of enrollments).
- The organizer of a tournament must be a player of the club.
- The number of rounds must be defined when creating a tournament and this number will be part of the tournament information.
- Two leagues cannot have the same name.
- The stock of a gift must be always ≥ 0 .
- Players cannot get gifts if they have not enough rating points.
- Players cannot get gifts if the stock of the gift is lower than the number of gifts.



Requested information

1. Write the SQL statements required to create the relational schema, including all primary and foreign key integrity constraints and CHECK constraints.
2. Write the SQL statement to insert records:
 - 2.1. Insert a minimum of 10 records in each table.
 - 2.2. Insert records into the tables so that records are obtained as a result when executing the next queries.
3. Specify the following queries in SQL
 - 3.1. Query 1: Obtain the average number of tournaments organized for each active player in each club.
 - 3.2. Query 2: Get the name of the clubs, and, for each title, the title name and the number of active players with this title in the club, sorting the result by club name in alphabetical descending order.
 - 3.3. Query 3: Obtain the number of gifts bought per player since 2021 for each type of gift.
 - 3.4. Query 4: Calculate the number of games won by each player in each tournament with white pieces. List the name of the tournament, the date, the name of the team and the number of games won by each team, sorting the results by tournament's date in ascending order and by number of games won with white pieces in descending order.
4. Specify the following trigger in SQL
 - 4.1. When creating a club, the first one has a code 1, and the rest are assigned the following sequential number (i.e. the trigger calculates the code and assigns the code to each new club so that it is a sequential number).
 - 4.2. Trigger: When registering a game, check that the black and white players are part of the club that organized (and they haven't left the club).
 - 4.3. Trigger: Update the rating of the players' game when the result of the game is registered.
 - 4.4. Trigger: When a player buys a gift, update the player's rating and the gift's stock. Raise an error in case the player doesn't have enough rating or the stock of the gift is 0.

Instructions, date

This project must be handed in on November 1st, 2024, via the "Aula Global" platform (by one of the group members only). The document must be uploaded in ".zip" or ".rar" format, the compressed file should include:

1. File in txt format with the script of creation of tables, with the restrictions of PK, AK and CHECK.
2. File in txt format with the script for the deletion of the tables.
3. File in txt format with the script for insertion of the records.
4. File in txt format with the script for the creation of FK constraints.
5. File in txt format with the script for the queries and triggers.